

□ PROGRAMMING FUNDAMENTALS - LAB

Assignment:

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Program: BS(AI)

Section: BAI-1A

Obj. 1: PROBLEM ANALYSIS CHART

Problem #1: Student Details Record:

DATA	OUTPUT
Student ID Student Name Subject 1 Marks Subject 2 Marks Subject 3 Marks Max. marks = 100 Marks per subject	Student ID Student Name Obtained Marks Percentage
PROCESSING	ALTERNATE SOLUTION
Obtained = S1 + S2 + S3 Maximum = $100 \times 3 = 300$ Percentage = $(\text{Obtained}/300) \times 100$ Display the required information	Use a loop for subject input Use a list and sum Use a function for calculations

Problem #2: Area of a Rectangle:

DATA	OUTPUT
Length Width	Area of Rectangle
PROCESSING	ALTERNATE SOLUTIONS
Input length and width Area = Length and width Display the calculated area	Use a function to calculate area Use length $\times$ width directly while displaying the result.

Problem #3: Celsius to Fahrenheit:

DATA	OUTPUT
Temperature in Celsius	Temperature in Fahrenheit
PROCESSING	ALTERNATE SOLUTION
Input Celsius temperature Fahrenheit = $(C \times 9/5) + 32$ Calculate Fahrenheit Display Fahrenheit	Use a function to perform the conversion

Obj. 2: PSEUDOCODE

Problem #1: Find the maximum number in any of three variables:

```
                START
INPUT A
INPUT B
INPUT C

IF A >= B AND A >= C THEN
DISPLAY "Maximum number is:", A
ELSE IF B >= A AND B >= C THEN
DISPLAY "Maximum number is:", B
    ELSE
DISPLAY "Maximum number is:", C
    END IF

END
```

Problem #2: Add three variables as input and add them without using the + operator:

```
                START
INPUT A
INPUT B
INPUT C

SUM = A - (-B) - (-C)

DISPLAY "Sum is:", SUM

END
```

Problem #3: Create a Small calculator for Addition and Substraction operations:

```

START
INPUT A
INPUT OPERATOR
INPUT B

IF OPERATOR = "+" THEN
RESULT = A + B
DISPLAY "Result is:", RESULT

ELSE IF OPERATOR = "-" THEN
RESULT = A - B
DISPLAY "Result is:", RESULT

ELSE
DISPLAY "Invalid operator"

END IF

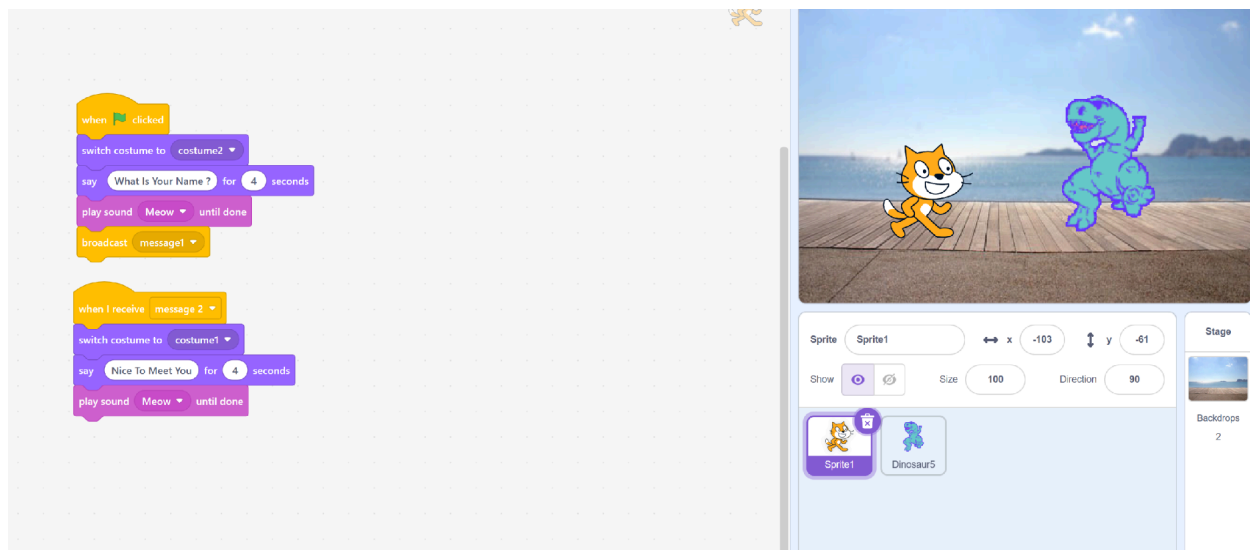
END

```

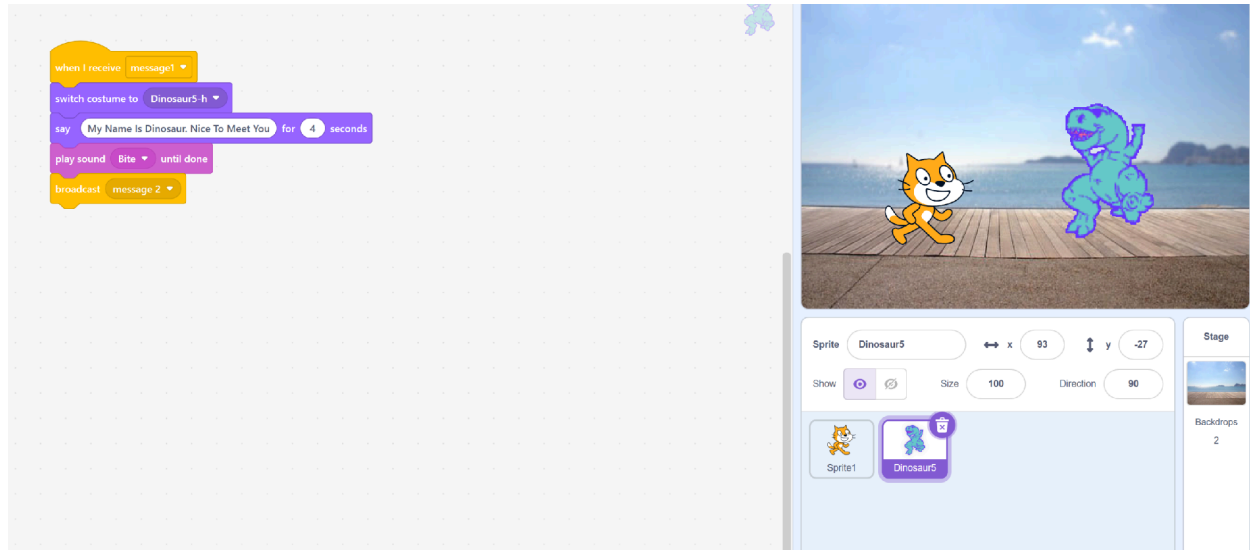
Obj. 3: SCRATCH

Problem #1: Create a Scratch program that demonstrates a conversation between two sprites using the Broadcast and When I Receive blocks:

SPRIT 1: CAT:



SPRIT 2: DINOSAUR:



PROBLEM #7: Create a Scratch program in which the sprite introduces itself when the green flag is clicked:

when green flag clicked

say Hello! for 4 seconds

say My Name Is Cat. for 4 seconds

switch costume to costume2

move 20 steps

say Welcome! Nice To Meet You. for 4 seconds

Sprite1

Dinosaur

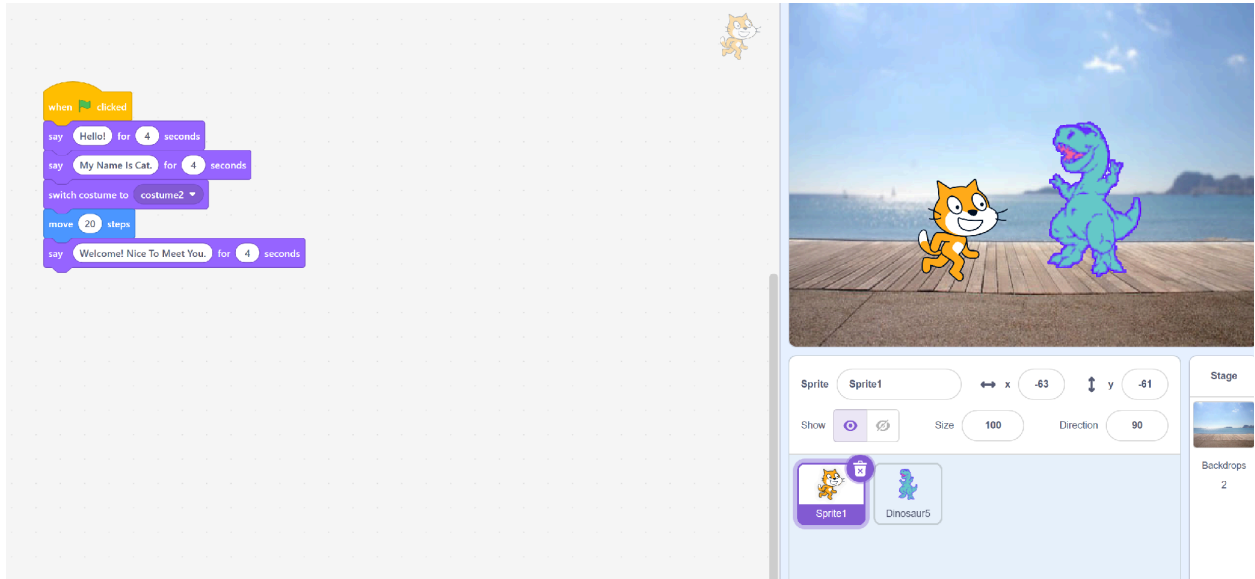
Sprite1

x -63y -61

ShowSize 100Direction 90

Stage

Backdrops2



The image shows a Scratch project interface. On the left, a script is built on a grid background. It starts with a 'when green flag clicked' event block, followed by a 'say Hello! for 4 seconds' block, a 'say My Name Is Cat. for 4 seconds' block, a 'switch costume to costume2' block, a 'move 20 steps' block, and finally a 'say Welcome! Nice To Meet You. for 4 seconds' block. On the right, the stage view shows a cartoon cat sprite (Sprite1) and a blue dinosaur sprite (Dinosaur) on a wooden pier. The cat is on the left, and the dinosaur is on the right. The background is a beach scene with a blue sky, ocean, and distant hills. The stage view also includes a 'Sprite' panel with 'Sprite1' and 'Dinosaur' options, and a 'Stage' panel with 'Backdrops2'.